

Project Proposal

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1. The main goal of our project is to evaluate, observe and discuss ties between football players and teams based on dataset provided by real information from TransferMarket. Nowadays, this quick way to transfer a player during mercato periods has significantly grown and the importance of this factor should be a serious issue for the teams. The dynamic of the transfer market could be studied over the last 25 years, making some key points of discussion and how it is going to change in the future. Using network science, we can analyze player mobility and club interdependence, offering at the same time a data-drive perspective.
2. The first network that should be extracted from the dataset is the bipartite network (player-club) which connects players to their respective clubs; it is built from contract and appearance data. This network will help us notice degree distributions(player per club, clubs per player, identifying highly connected clubs). This graph could also be weighted based on the number of appearances or minutes played for that club(this will be defined later in order to fulfill challenge grading criteria).

The second network defined for this study is the club-club transfer network. This is a directed, weighted network which shows the movements across the season. Edges could indicate transfers and the weight would be the value of players transferred. This should help us identify the major clubs which often transfer and notice inter-league movements.

3. The dataset that we are going to use is provided by TransferMarkt(a real website which contains all realistic details about a transfer, club and players). Below is the link from where we got the dataset that should be preprocessed before comes into play in our network configuration(get rid of few entries, checking if it is correct): <https://www.kaggle.com/datasets/davidcariboo/player-scores>
It is definitely related to our topic because the dataset has all the details that we need in order to discuss and have results on the problem approached.

4. Table "Who did what?"

Task	Time	Who?
Preprocessing DataSet	1 week	Ștefan
Create Networks	1 week	Both
Metrics and Discussion	1 week	Eduard
Conclusion and Presentation	1 week	Both
Finish the report	a few days	Both